Two-view Geometry (II)

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Questions?

- HW1 due on Wednesday
- HW2 out next week
  - More coding-heavy

Recap: essential matrix

Define the **Essential matrix**

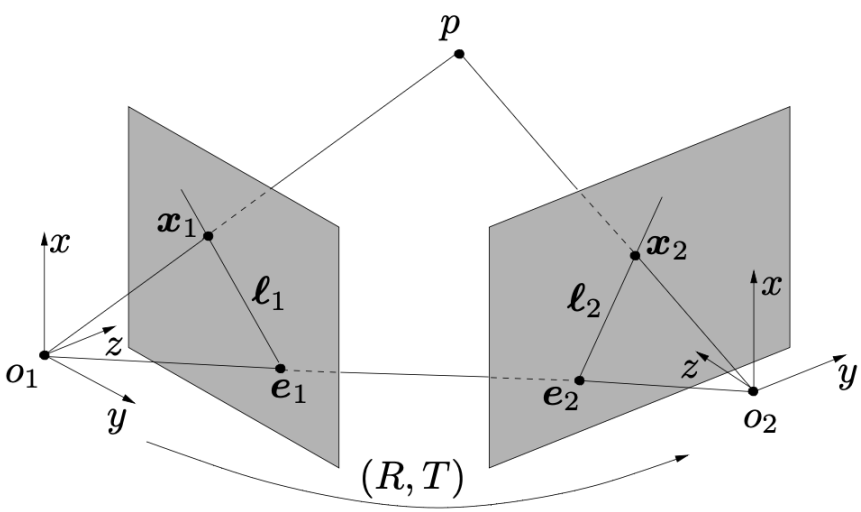
$$\mathbf{E} \triangleq [\mathbf{T}]_\times \mathbf{R} \in \mathbb{R}^{3\times 3}.$$

- E always has a rank of 2
- E has two identical singular values

Recap: Epipolar constraint

In the same coordinate system, three things lie on the same plane:

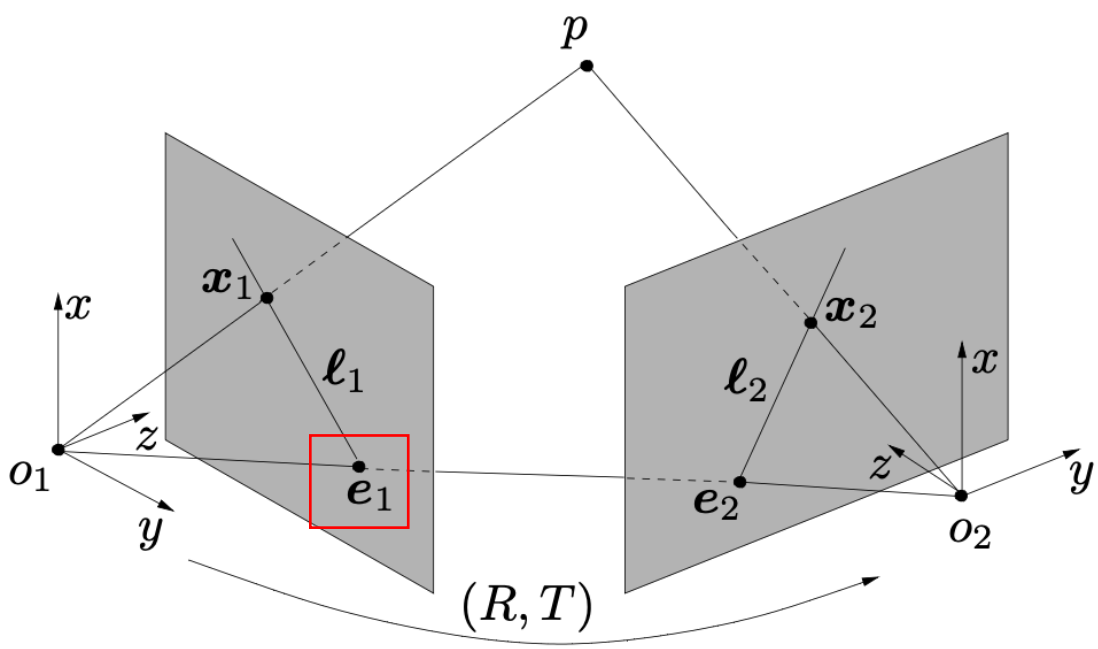
1. Ray of p in the second view (x2)
2. Ray of p in the second view, written in the first frame (R x2)
3. Vector O2 to O1, written in the first frame (T)



$$\boxed{\boldsymbol{x}_2^T \widehat{T} R \boldsymbol{x}_1 = 0}$$

Epipole

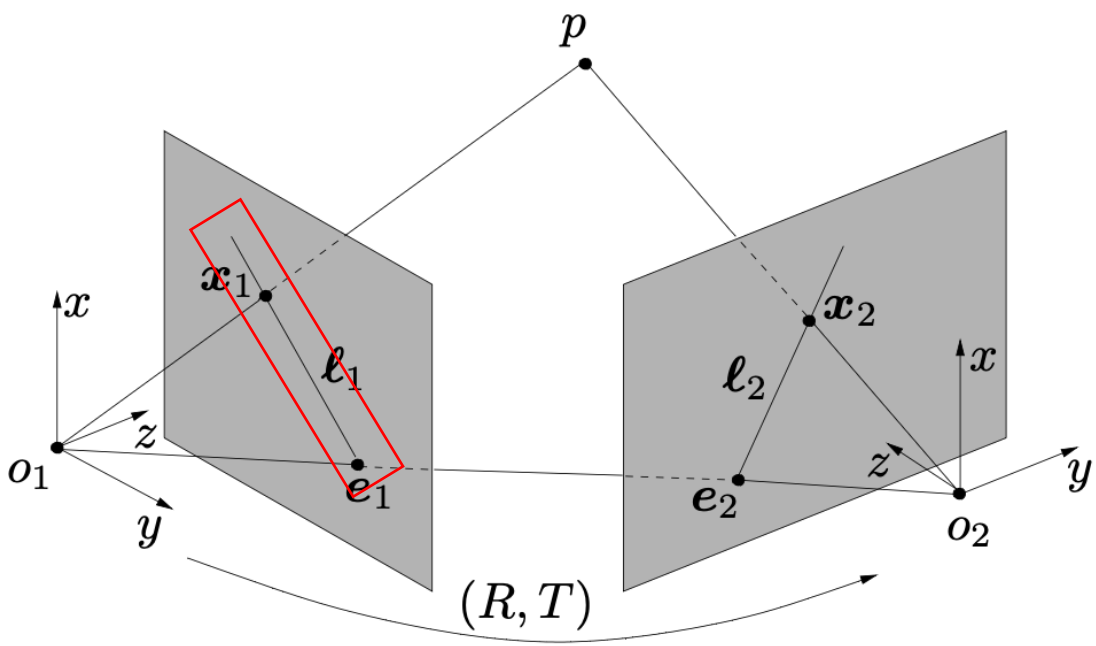
"The projection of the other camera's origin in the current image"



Also projection of the the direction of translation

Epipolar Lines

"Line connecting a pixel to the Epipole"



Recap: Epipolar lines

The essential matrix maps a point in view 1 to an "epipolar line" in view 2

Given a point $\boldsymbol{x}_1$ in image 1, the corresponding epipolar line in image 2 is

$$\boldsymbol{\ell}_2 \sim \mathbf{E}\boldsymbol{x}_1.$$

Given $\boldsymbol{x}_2$ in image 2, the corresponding epipolar line in image 1 is

$$\boldsymbol{\ell}_1 \sim \mathbf{E}^\top \boldsymbol{x}_2.$$

Estimate R and T from E

- Yes, but up to a scale in translation

SVD again!

Closest Essential matrix in Frobenius norm

$$\mathbf{E} = \mathbf{U}\,\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{V}^\top, \qquad \boxed{\sigma = \frac{\lambda_1 + \lambda_2}{2}.}$$

Because two identical singular values can also be set to all 1

Let $\mathbf{E} = \mathbf{U}\,\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{V}^\top$.

Then a translation direction is (up to sign)

$$\hat{\mathbf{T}} \sim \mathbf{U}(:, 3).$$

Last column of U is a translation

1) The last column of $\mathbf{U}$ spans $\mathrm{Null}(\mathbf{E}^\top)$

Transpose the SVD:

$$\mathbf{E}^\top = \mathbf{V}\,\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{U}^\top.$$

If $\mathbf{E}^\top \boldsymbol{x} = \mathbf{0}$ then

$$\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{U}^\top \boldsymbol{x} = \mathbf{0} \;\Rightarrow\; \mathbf{U}^\top \boldsymbol{x} \in \mathrm{span}(\boldsymbol{e}_3) \;\Rightarrow\; \boldsymbol{x} \in \mathrm{span}(\boldsymbol{u}_3).$$

2) $\mathbf{T} \in \mathrm{Null}(\mathbf{E}^\top)$

Using $\mathbf{E} = [\mathbf{T}]_\times \mathbf{R}$,

$$\mathbf{E}^\top \mathbf{T} = \mathbf{R}^\top [\mathbf{T}]_\times^\top \mathbf{T} = -\mathbf{R}^\top ([\mathbf{T}]_\times \mathbf{T}) = -\mathbf{R}^\top (\mathbf{T} \times \mathbf{T}) = \mathbf{0}.$$

Rotations

Let $\mathbf{E} = \mathbf{U}\,\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{V}^\top$.
Define

$$\mathbf{W} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

candidate rotations are

$$\mathbf{R}_1 = \mathbf{U}\mathbf{W}\mathbf{V}^\top, \qquad \mathbf{R}_2 = \mathbf{U}\mathbf{W}^\top\mathbf{V}^\top.$$

Important detail

Ensure $\det(\mathbf{R}) = +1$; if $\det(\mathbf{U}\mathbf{V}^\top) = -1$, flip the sign of one column of $\mathbf{U}$ or $\mathbf{V}$.

How many solutions?

From decomposition you typically get **four** candidates:

$$(\mathbf{R}_1,\ +\hat{\mathbf{T}}),\ (\mathbf{R}_1,\ -\hat{\mathbf{T}}),\ (\mathbf{R}_2,\ +\hat{\mathbf{T}}),\ (\mathbf{R}_2,\ -\hat{\mathbf{T}}).$$

Often described as two rotation choices (a *twisted pair*) and an unknown translation sign.

Cheirality test

Given a candidate $(\mathbf{R}, \mathbf{T})$:

1. Triangulate a few correspondences to get 3D points $\boldsymbol{X}$.
2. Check depths are positive in both cameras:

$$z_1 > 0, \qquad z_2 > 0.$$

Pick the candidate where the largest number of points satisfy positive depth.

Solve essential matrix given point pairs

Each correspondence provides

$$\boldsymbol{x}_2^\top \mathbf{E} \boldsymbol{x}_1 = 0.$$

Linear in entries of $\mathbf{E}$

Define the vectorized essential matrix (row-major stacking)

$$\mathbf{E}_s \triangleq \begin{bmatrix} E_{11} & E_{12} & E_{13} & E_{21} & E_{22} & E_{23} & E_{31} & E_{32} & E_{33} \end{bmatrix}^\top \in \mathbb{R}^9.$$

Then one correspondence yields a linear equation

$$(u_1 u_2,\ v_1 u_2,\ u_2,\ u_1 v_2,\ v_1 v_2,\ v_2,\ u_1,\ v_1,\ 1)\, \mathbf{E}_s = 0.$$

Solve with RANSAC

Epipolar error

Each correspondence provides

$$\boxed{\boldsymbol{x}_2^\top \mathbf{E}}\boldsymbol{x}_1 = 0.$$

Epipolar line for x2 in view1

Eight Point Algorithm

- Like solving homography

$$\underbrace{(u_1u_2,\ v_1u_2,\ u_2,\ u_1v_2,\ v_1v_2,\ v_2,\ u_1,\ v_1,\ 1)}_{\boldsymbol{a}^\top = (\boldsymbol{x}_2 \otimes \boldsymbol{x}_1)^\top} \mathbf{E}_s = 0. \qquad \chi = \begin{bmatrix} \boldsymbol{a}_1^\top \\ \vdots \\ \boldsymbol{a}_n^\top \end{bmatrix} \in \mathbb{R}^{n \times 9}.$$

- compute SVD $\chi = \mathbf{U}_\chi \Sigma_\chi \mathbf{V}_\chi^\top$,
- take $\mathbf{E}_s$ as the last column of $\mathbf{V}_\chi$.

Correcting essential matrix

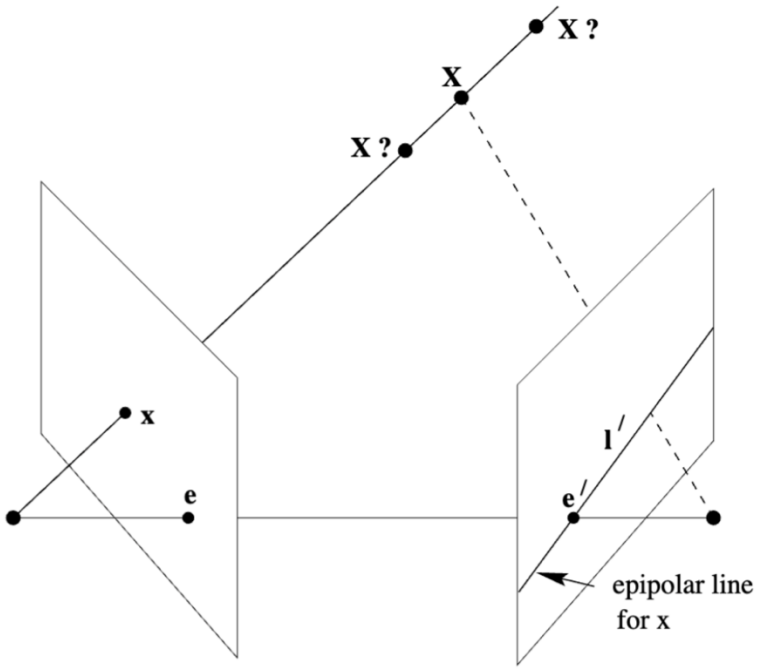
After the linear step, $\mathbf{F}$ will not exactly satisfy the essential constraints.
Here $\mathbf{F} = \mathrm{unvec}(\mathbf{E}_s)$ is the 3×3 matrix reshaped from the 9-vector $\mathbf{E}_s$ above.
Compute SVD:

$$\mathbf{F} = \mathbf{U}\,\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)\,\mathbf{V}^\top, \quad \lambda_1 \geq \lambda_2 \geq \lambda_3.$$

Closest Essential matrix in Frobenius norm

$$\mathbf{E} = \mathbf{U}\,\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{V}^\top, \qquad \sigma = \frac{\lambda_1 + \lambda_2}{2}.$$

Reconstruction?



The essential matrix gives us the pose, but not where the points are.
We still need correspondence (along the epipolar line)

Triangulation

With $K = I$ and known pose up to translation scale γ:

$$\lambda_2 \boldsymbol{x}_2 = \lambda_1 \mathbf{R} \boldsymbol{x}_1 + \gamma \, \mathbf{T}.$$

Unknowns: $(\lambda_1, \lambda_2, \gamma)$.

Solve as a linear system

Rearrange as

$$\underbrace{\begin{bmatrix} \underbrace{\mathbf{R}\boldsymbol{x}_1}_{3\times 1} & \underbrace{-\boldsymbol{x}_2}_{3\times 1} & \underbrace{\mathbf{T}}_{3\times 1} \end{bmatrix}}_{3\times 3} \underbrace{\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \gamma \end{bmatrix}}_{3\times 1} = \underbrace{\mathbf{0}}_{3\times 1}.$$

Solve in least squares (e.g., SVD) for each point, or use standard triangulation.

Known translation

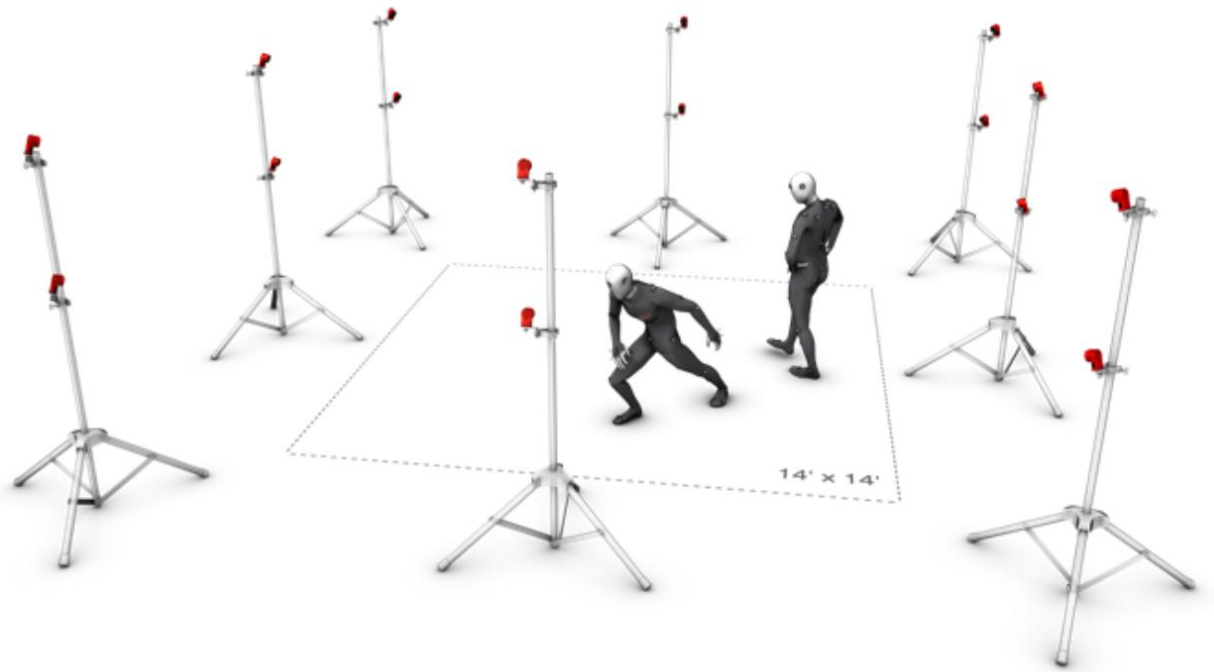
If $\mathbf{R}$ and the full translation vector $\mathbf{T}$ are known (no unknown scale γ), then

$$\lambda_2 \boldsymbol{x}_2 = \lambda_1 \mathbf{R} \boldsymbol{x}_1 + \mathbf{T}.$$

Rearrange into a 3×2 linear system in the two depths:

$$\underbrace{\begin{bmatrix} \boldsymbol{x}_2 & -\mathbf{R}\boldsymbol{x}_1 \end{bmatrix}}_{3\times 2} \underbrace{\begin{bmatrix} \lambda_2 \\ \lambda_1 \end{bmatrix}}_{2\times 1} = \underbrace{\mathbf{T}}_{3\times 1}.$$

Motion capture system



What problems are we solving?

$$\mu_i \, \boldsymbol{q}_i = \mathbf{T} + \lambda_i \, \mathbf{R} \boldsymbol{p}_i$$

Known values	Problem	To solve
$\{\mu_i, \lambda_i, \boldsymbol{p}_i, \boldsymbol{q}_i\}$	Procrustes / Kabsch	$(\mathbf{R}, \mathbf{T})$
$\{\lambda_i, \boldsymbol{p}_i, \boldsymbol{q}_i\}$	PnP (3D–2D bearings)	$(\mathbf{R}, \mathbf{T})$ $(+\{\mu_i\})$
$\{\boldsymbol{p}_i, \boldsymbol{q}_i\}$	two-view SfM (estimate $\mathbf{E}$)	$\mathbf{E} \Rightarrow (\mathbf{R}, \pm\hat{\mathbf{T}}),\ \{\lambda_i, \mu_i\}$

Special case 1: homography

Rotation-only:

If $\mathbf{T} = 0$ then

$$\mathbf{E} = [\mathbf{T}]_\times \mathbf{R} = \mathbf{0}.$$

Translation direction is unobservable.
In this case the relationship is a homography on the sphere / image:

$$\boldsymbol{x}_2 \sim \mathbf{R}\boldsymbol{x}_1.$$

Planar surface:

If all points lie on a plane, correspondences can be explained by a homography:

$$\boldsymbol{x}_2 \sim \mathbf{H}\boldsymbol{x}_1.$$

Multiple solutions correspond to the same essential matrix

Special case 2: small parallax

- The location of a point does not change much between two views
  - Points are too far away
  - Translation is too small

"Structure-from-motion" -> no motion, no structure

Fundamental Matrix

Definition

The **Fundamental matrix** $\mathbf{F} \in \mathbb{R}^{3\times 3}$ satisfies

$$\tilde{\boldsymbol{x}}_2^\top \, \mathbf{F} \, \tilde{\boldsymbol{x}}_1 = 0 \qquad \text{(uncalibrated / pixels).}$$

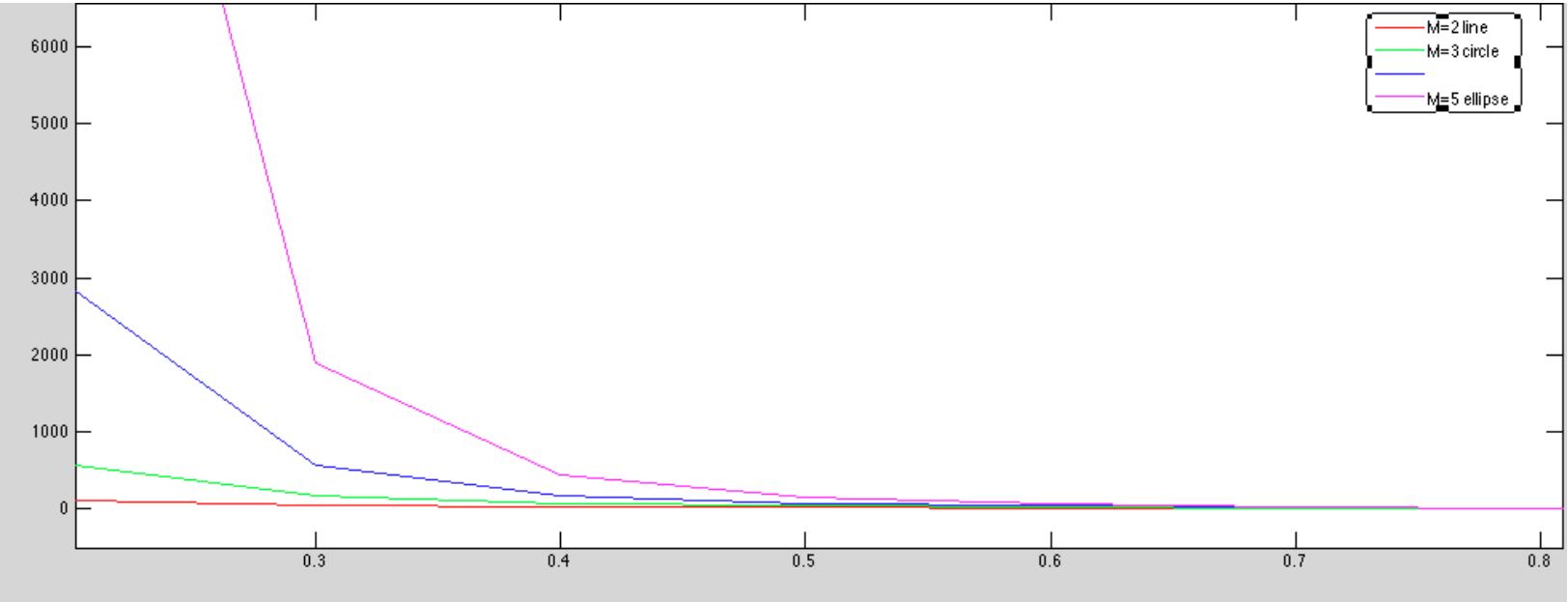
Key relationship

$$\boxed{\mathbf{F} \sim \mathbf{K}_2^{-\top} \, \mathbf{E} \, \mathbf{K}_1^{-1}}$$

If intrinsics are known, you can convert $\mathbf{F} \to \mathbf{E}$ (up to scale), then decompose to pose.

We usually assume the intrinsics are known and calibrated.

RANSAC?



The naive linear method uses 8 points to solve essential matrix

5-Point algorithm

- Fewer points, more common in pose estimation
- 5 pairs of points, and
- Up-to 10 solutions

```
E, inlier_mask = cv2.findEssentialMat(
    pts1, pts2, K,
    method=cv2.RANSAC,
    prob=0.999,
    threshold=1.0  # reprojection threshold in pixels (for RANSAC inlier test)
)
```

Summary

- Core constraint: $\boldsymbol{x}_2^\top \mathbf{E} \boldsymbol{x}_1 = 0$ with $\mathbf{E} = [\mathbf{T}]_\times \mathbf{R}$.
- $\mathbf{E}$: rank 2 with two equal nonzero singular values.
- 8-point algorithm: linear estimate $\rightarrow$ project to Essential space.
- Decomposition gives multiple candidates; cheirality chooses the correct one.
- Depth/structure recovered by triangulation, up to global scale.

Tutorial on Colab